

Project Proposal

1U Cosmic Gamma-Ray Burst Nanosatellite Observatory

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1 Overview

The Low-Earth Orbit Gamma-Ray Mapping project is a space research initiative whose main aim is to design a highly constrained 1U Satellite. The primary engineering goal is to design and fabricate a nano satellite capable of collecting high energy waves in a $10\times 10\times 11.3$ cm volume.

The satellite converts the incoming high energy radiations into electrical signals. These signals are then compiled into small data packets utilizing a thin plate crystal array (Thallium Doped Cesium Iodide) which is connected to a high sensitive photon sensor. Then these electrical signals are sent to a micro controller, which converts them into telemetry low bandwidth data which is then streamed back to the station. The crystal will be facing opposite to the Earth, using a small magnet and the Earth's magnetic field.

The Satellite will be using an open sourced tracking network to automatically collect and upload its science data to the web for public access. To securely send it to its orbit, the hardware and software are designed to be strictly in accordance with the NASA requirements.

2 Regulatory Pipeline & Legal Strategy

To secure a lift from NASA and the legal right to transmit data back to ground, we have to keep in mind the following documents:

2.1 IARU Frequency Coordination

This application will be submitted to the International Amateur Radio Union (IARU), once a Stony Brook faculty advisor signs it, and once the documentation for the specific frequency band we are applying for is complete, we just have to wait for approval. It might take around 6-12 months for the approval so this document needs to be sent in as soon as possible.

This will give us frequency band (435-438 MHz) for sending and receiving data to and from the satellite. Students and Professors all over the world can stream our network for free because the IARU integrates the open sourced network I talked about (SatNOGS) with the frequency that we will be allotted. This is more than enough bandwidth for binary data and we can afford an antenna that is around 16 cm long, hence, the 435-438 MHz work for our use.

2.2 NASA CSLI Flight Proposal

The mission, goals and the designs will be documented and in coordination with the faculty advisor, the team will have to submit a technical proposal to the NASA CubeSat Launch Initiative (CSLI) by November, and when, if the project proposal is selected in March, we will have 1 to 2 years to build the actual satellite and all other stations.

2.3 FCC Part 5 Experimental Space Station License

The physical structural materials (Aluminum 6061-T6) and maximum allowable mass boundaries (1.33 kg) are structurally locked. The faculty trustee will log into the FCC CORES portal to submit **FCC Form 442**. To pass mandatory range safety reviews, the finalized CAD models will be executed through NASA's free **Debris Assessment Software (DAS)** to generate an Orbital Debris Assessment Report (ODAR). This step mathematically proves the satellite will achieve complete atmospheric burn-up within 5 years of mission decommissioning.

The physical structural materials ($Al - 6061 - T6$) and max allowable mass and boundaries ($m = 1.33\text{ kg}$

$100 \times 100 \times 113.5\text{ mm}$) are set by NASA CSLik Program. The faculty advisor(s) will have to log in to the FCC CORES Portal to submit **FCC Form 442**. To pass mandatory safety reviews, the finalized CAD Models will be executed through the NASA's free **Debris Assessment Software (DAS)** to generate an Orbital Debris Assessment Report (ODAR).

3 Threshold of Detection

3.1 Detection Threshold & Fluence Sensitivity

The detection is primarily optimized for soft gamma and hard X rays. The sensitivity of the Thallium-doped Cesium Iodide is capable of yielding a Fluence Detection Threshold of 10^{-6} erg/cm^2 over the duration of the event. The crystal emits almost about 54 Green photons per keV of energy. ($\sim 550\text{nm}$)

In order to isolate the soft gamma radiations and hard X rays from background noise, the flight computer (STM Chipset -Stage 3) applies a hard-coded 5σ software filter over a continuous rolling window. (First it is going to record the ambient space noise from the sensor, and then take the average and the standard deviation of that data. Then it will setup a trigger point, this cancels out the noise from the useable data.)

$$Trigger = Mean + (5 * \sigma)$$

3.2 Annual Detection

Operating in Low Earth Orbit (LEO) at a nominal altitude of $\sim 420\text{ km}$ restricts the sensor's instantaneous field of view, as the physical mass of the Earth blocks around half of the sky. Based on the known historical distribution of energy, the satellite is projected to capture 10 to 20 soft gamma-ray bursts per year.

3.3 Energy and Long/Short G Rays

- The satellite has been tuned to detect Soft Gamma rays or Hard X-rays, mainly ranging from 30 keV to 300 keV. This is mainly due to a hardware limitation: the crystal that we are using has a thickness of 3 mm. In order to detect stronger waves, the thickness would have to be increased, which our design cannot afford. The efficiency for this crystal would be around 84.5%.

$$\eta = (Y) \cdot (E_{\text{Photon}})$$

As the yield of the crystal is 54 photons per keV (producing 54 photons per every 1,000 eV). Hence, finding the energy efficiency of the green photons emitted:

$$\eta = \left(54 \frac{\text{photons}}{\text{keV}} \right) \cdot \left(\frac{(6.63 \times 10^{-34} \text{ J} \cdot \text{s}) \cdot (3 \times 10^8 \text{ m/s})}{(550 \times 10^{-9} \text{ m}) \cdot (1.6 \times 10^{-16} \text{ J/keV})} \right)$$

$$\eta \approx 12.15\%$$

Hence, from this it is showing that the crystal has an energy efficiency of about 12.15 percent for a specific range of gamma radiations, while from Beer-Lambert's Law of Radiation, the physical tracking absorption efficiency is:

$$\alpha = 1 - \frac{I}{I_0} = 1 - e^{-\mu x}$$

Where μ is the linear attenuation coefficient of the material. For a Thallium-doped Cesium Iodide, this coefficient comes out as 6.21 cm^{-1} for 100 keV, and x is the thickness of the crystal. Hence, the equation comes out to be:

$$\alpha = 1 - e^{-6.21 \text{ cm}^{-1} \cdot 0.3 \text{ cm}}$$

$$\alpha = 84.5\%$$

- In order to differentiate between soft and some hard gamma rays, the software tracks the time-domain density to identify which gamma ray burst released the most photons over its full duration. When a gamma-ray burst hits the crystal, the sensor sends in data to the filter on Stage 2. The filter will be sending in pulses to the STM Chipset on Stage 3. Each individual pulse triggers a background interrupt that logs a precise time window. If we have more of these pulses incoming, they will populate a longer timeline, where the chipset will recognize it as a longer duration event.

$$1^{\text{st}} \text{ Pulse} = 0.00004 \text{ seconds}$$

$$2^{\text{nd}} \text{ Pulse} = 0.00007 \text{ seconds}$$

$$3^{\text{rd}} \text{ Pulse} = 0.00011 \text{ seconds}$$

Figure 1: Example plot demonstrating a short burst timeline, which could last anywhere from 1 to 5 seconds, contrasted against a longer burst timeline that could last up to around 40 seconds to a minute.

3.4 Volume and Aperture Fraction

The centred 50mm by 50mm by 3mm crystal that we are using is only fixed at a single direction. The satellite, itself has no space for any accomadation of actuation motors, hence the crystal acts as an open field sensor intercepting the space radiation coming towards the Earth.

3.5 Direction

The centered $50 \times 50 \times 3$ mm scintillation crystal acts as volumetric collector with zero directional tracking capabilities. The crystal’s 3 mm thickness will be able to capture approximately **84.5% of all available incident GRBs** that reach the spacecraft aperture window within our 30–300 keV detection threshold.

The core institutional value of this project is that it is 100% Open Source and its data will be released to the public using the SatNOGS platform.

3.6 Mapping

We designed the satellite’s data output to be very lightweight, creating simple binary text files that contain our raw microsecond timestamps. This formatting makes it straightforward to cross-reference our timeline with data from global networks like the LIGO-Virgo-KAGRA collaboration, which track deep-space gravitational waves.

When a ground-based observatory records a spacetime ripple from a potential binary star merger, researchers can check our open-source log to see if our crystal registered an event at that same time. Matching our gamma-ray timestamps with their gravitational wave data helps provide an independent dataset to check for a simultaneous electromagnetic flash, supporting the study of multi-messenger cosmic events.

4 Design

The structural envelope of the Low-Earth Orbit Gamma-Ray Mapping (LEO-GRM) space station utilizes a highly constrained 1U CubeSat form factor measuring exactly $100.0 \times 100.0 \times 113.5$ mm [4.1]. The internal payload and logic bus are distributed across a uniform vertical column consisting of 6 distinct internal board decks.

To secure launch manifest selection through the NASA CubeSat Launch Initiative (CSLI), the structural and digital architectures are engineered in strict accordance with the NASA Launch Services Program (LSP) safety requirements:

- **Chassis Material Constraints:** The primary physical cage skeleton must be machined from a solid block of Aluminum 6061-T6 to survive structural flight vibration coefficients [4.1].
- **Orbital Startup Safety Lock:** The flight computer firmware enforces an un-bypassable, hardcoded 45-minute silent countdown delay timer immediately following deployment switch release from the rocket canister [4.1]. During this initial window, the satellite must remain completely dormant with all transmitter modules isolated from power to prevent RF interference with the launch vehicle [4.1].
- **Emergency Power Disconnect:** A dual-rail hardware deployment kill-switch is integrated into the structural rail faces to permanently cut power to every single internal component in the event of an unfused electrical failure.
- **Asymmetrical Link Encryption:** In compliance with international space treaties and range safety limits, the Earth-to-space telecommand uplink channel implements a secure

rolling-code cryptographic signature to protect steering nodes from unauthorized ground hijacking, while the space-to-Earth scientific telemetry downlink remains 100% unencrypted and open-source.

Figure 2: Isometric 3D CAD rendering of the LEO-GRM 1U CubeSat mechanical envelope. Image should highlight the solid Aluminum 6061-T6 chassis rails, deployment kill-switch placement on the deployment slides, flush-mounted exterior solar arrays, and the top-tier payload entry window aperture.

4.1 Brief Design Overview

The internal stack architecture completely replaces custom structural frame plates with native Printed Circuit Boards (PCBs) acting as the physical load-bearing tiers. Every individual stage utilizes an identical 95.0×95.0 mm square footprint outline to maintain axial center-of-mass symmetry within a < 2.0 cm spherical margin [4.1]. The boards are locked onto vertical corner support columns using M3 stainless steel threaded rods, with the vertical clearance spacing between decks variable based on component heights.

To minimize overall vehicle manufacturing costs, external solar harvesting arrays are constructed by surface-mounting space-grade Gallium Arsenide (GaAs) solar cells directly onto low-cost, open-source custom exterior PCBs mounted flush to five outer facets of the chassis shell.

Electrical interconnectivity and inter-board signal routing across the vertical stack are achieved through dual-row 40-pin perimeter headers (2.54mm pitch stacking columns) positioned on the left and right borders of every board. This creates a parallel backbone bus; for example, if a +5.0V power potential is asserted on a specific header pin on the power layer, this potential is distributed uniformly up and down the vertical stack column, allowing any board layer to tap into the bus without custom wiring harnesses.

Figure 3: Cross-sectional layer stack architecture diagram showing the uniform 95 mm x 95 mm PCB column profile. Graphic must visually document the dual-row 40-pin perimeter pass-through stacking header rails flanking the left and right borders, proving parallel inter-tier rail routing distribution.

4.2 Stage 1 – Science Payload Deck

Stage 1 is the primary data collection tier positioned at the absolute roof of the internal stack, directly beneath the optical aperture window facing deep space [4.1]. The core instrument features a flat $50 \times 50 \times 3$ mm Thallium-doped Cesium Iodide [CsI(Tl)] scintillation crystal plate wrapped tightly in 0.1 mm Enhanced Specular Reflector (ESR) mirror foil [4.1].

To keep the active crystal face pointing directly toward deep space without needing heavy mechanical motors, the spacecraft implements a Passive Magnetic Attitude Control (PMAC) loop. A permanent Neodymium bar magnet mounted on the internal structure co-aligns the satellite's Z-axis with the Earth's local geomagnetic field lines, while perpendicular high-permeability Mu-Metal

hysteresis rods damp out residual kinetic rotational spin, naturally orienting the roof window away from the Earth at high latitudes.

The crystal is optically coupled to a high-sensitivity Hamamatsu S13360-6050PE Silicon Photomultiplier (SiPM) array (JLCPCB Part: C42451657) to read the internal photon flashes. Because high-frequency switching noise on the main 40-pin perimeter stacking headers can easily corrupt the fragile millivolt-scale analog pulses, a dedicated 3-pin localized shielded jumper terminal is added to this stage to pass the signal out safely. On the sensor board layout, the high-voltage +53V bias supply pin is decoupled right at the pad through a 10nF 100V capacitor to absorb voltage ripples, while the signal anode pin routes across a straight, short 50-Ohm trace directly into a termination load resistor array to convert the ionization current into a clear voltage spike.

Figure 4: Exploded mechanical assembly sequence diagram of the Stage 1 payload sandwich array. Diagram must show the concentric layering starting from the base PCB, the centered S13360-6050PE SiPM array layout footprint, the 0.05 mm optical coupling gel interface, the 50x50x3mm CsI(Tl) crystal core, the 0.1mm ESR mirror foil jacket wrap, and the Aluminum 6061-T6 top retention bracket ring clamped with corner M3 screws.

4.3 Stage 2 – Front-End Signal Digitization

Stage 2 houses the open-source analog front-end pulse conditioning architecture of the Pico Gamma Spectrometer [4.1]. Placed 5.0 mm directly beneath Stage 1, this tier mates to the payload deck via the 3-pin shielded jumper cable [4.1]. Pin 1 establishes common ground, Pin 2 accepts the raw 2mV to 10mV analog ionization pulse, and Pin 3 drops the high-voltage +53V bias rail down into a local high-frequency C-L-C Pi-Filter circuit (100nF Cap $\rightarrow$ 600-Ohm Ferrite Bead $\rightarrow$ 10nF Cap) to clean the sensor power feed.

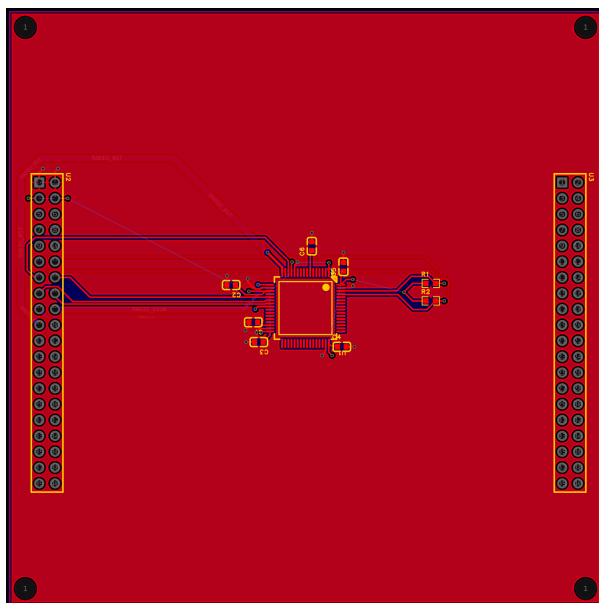
The raw pulse is processed by a high-speed, low-noise LTC6227HMS8 operational amplifier (LCSC Part: C2924151) powered by an ultra-quiet 3.3V linear LDO regulator [4.1]. The first stage converts the weak SiPM current into a voltage wave using a parallel 10k Ω resistor and 10pF static filtering capacitor loop to dump ambient background RF interference [4.1]. The second stage acts as a high-speed comparator that shapes the wave into a crisp 3.3V digital square pulse. This finished logic pulse travels down a bottom-layer 1x3 connector straight into the flight computer below.

Figure 5: Detailed electronic wiring schematic layout for the Stage 2 Pico Gamma Spectrometer board. Diagram should illustrate the +53V input line entering the C-L-C Pi-filter network, the input pulse routing directly into the LTC6227HMS8 inverting configuration, the 10pF low-pass RF suppression link, and the output trace dropping into the bottom-layer board-to-board logic junction.

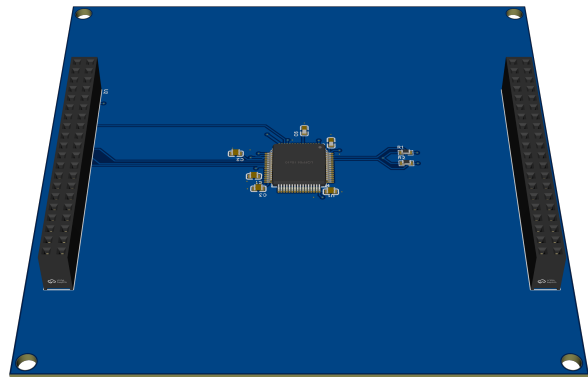
4.4 Stage 3 – Command & Data Handling (C&DH Flight Computer)

Stage 3 serves as the primary computational flight computer core. The board is centered around an STMicroelectronics STM32F411RET6 32-bit ARM Cortex-M4 microcontroller chip (LCSC Part: C61314) running at 100 MHz [4.1]. The 3.3V digital square pulse output from Stage 2 maps directly into STM32 Pin PA10, which is configured as a background hardware interrupt (EXTI10) to log radiation hits instantly with a 1-microsecond timestamp resolution.

The complete flight computer software engine is written natively in C++ to achieve rapid execution speeds. The firmware is responsible for executing the mandatory 45-minute post-deployment launch safety silence window, running the moving 5-Sigma statistical background calculation tracking loop, managing data packet transactions with our onboard storage blocks, and driving the antenna release logic. In the case of a catastrophic system error or a ground-directed command failure, the computer drops its logic pins to instantly cut the distribution voltages and freeze all active processes.



(a) C&DH Core 2D Layout Topology



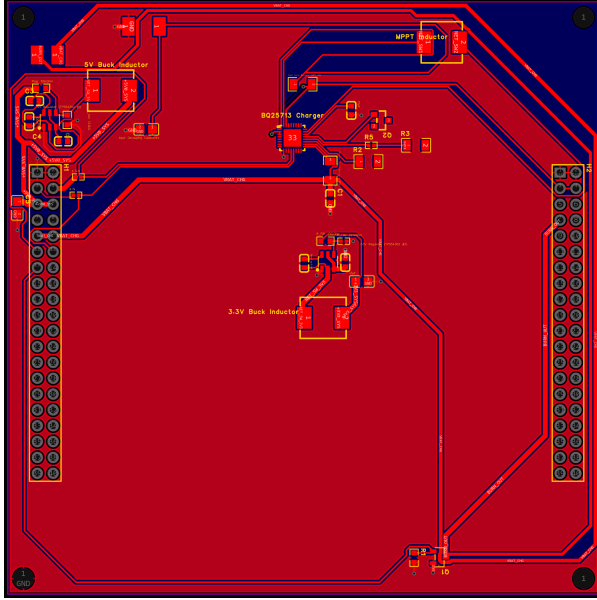
(b) 3D Logic Tier Clearance Model

Figure 6: Command and Data Handling (C&DH) structural subsystem layout configurations for the Stage 3 main flight computer. Panel (a) illustrates the dense routing geometry and escape via fanouts for the centered 64-pin STM32F411RET6 MCU microchip, including dual $4.7\text{k}\Omega$ pulling-resistors on the I2C1 bus lines (PB8/PB9). Panel (b) models the physical volumetric component spatial heights, demonstrating 100nF decoupling capacitors situated closely to the active VDD banks to defend the ARM Cortex-M4 silicon core against high orbital vibration profiles.

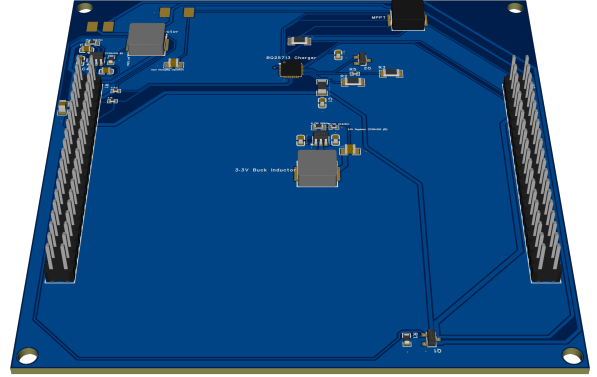
4.5 Stage 4 – Electrical Power System (EPS Power Manager)

Stage 4 functions as the central energy harvesting and power distribution hub for the entire satellite structure. This tier integrates a Texas Instruments BQ25713 Buck-Boost MPPT Solar Charging Controller to handle real-time power point optimization from the five external GaAs solar cell panels.

The board houses high-efficiency inductors, switching FETs, and linear voltage regulators capable of converting unstable solar cell inputs into regulated, flat +3.3V and +5.0V system logic rails. All power paths on the copper layers are widened to 1.0 mm traces to handle high-current transients safely without creating heat buildup or voltage sags under load.



(a) 2D Copper Trace Routing Layout



(b) 3D Structural Component Model

Figure 7: Electronic Power System (EPS) engineering layout views for the Stage 4 subsystem. The left plot (a) documents the high-density copper trace routing paths configured on a 4-layer stackup matrix. The right plot (b) illustrates the volumetric component clearance heights and physical positioning bounds for the Texas Instruments BQ25713 MPPT solar controller, power inductors, and perimeter pass-through stacking headers.

4.6 Stage 5 – Batteries UHF Transceiver Ballast

Stage 5 acts as the high-mass basement ballast floor positioned at the bottom of the CubeSat structure to keep the center of gravity low and stable [4.1]. This tier holds two 18650 Lithium Iron Phosphate (LiFePO_4) battery cells wired in series (2S1P configuration / nominal 6.4V) and secured inside ultra-rigid, through-hole Keystone 1042 metal retention clips to resist high launch forces.

This layer also hosts the Adafruit RFM95W UHF Transceiver module (433 MHz / Product ID 3073), which houses an internal onsemi AX5043 sub-GHz radio engine. The module encodes data streams into Gaussian Frequency Shift Keying (GFSK) telemetry frames at 9600 baud using AX.25 standard packet radio framing, routing its raw RF signal lines directly down into the adjacent stacking header pins to drive the antenna array.

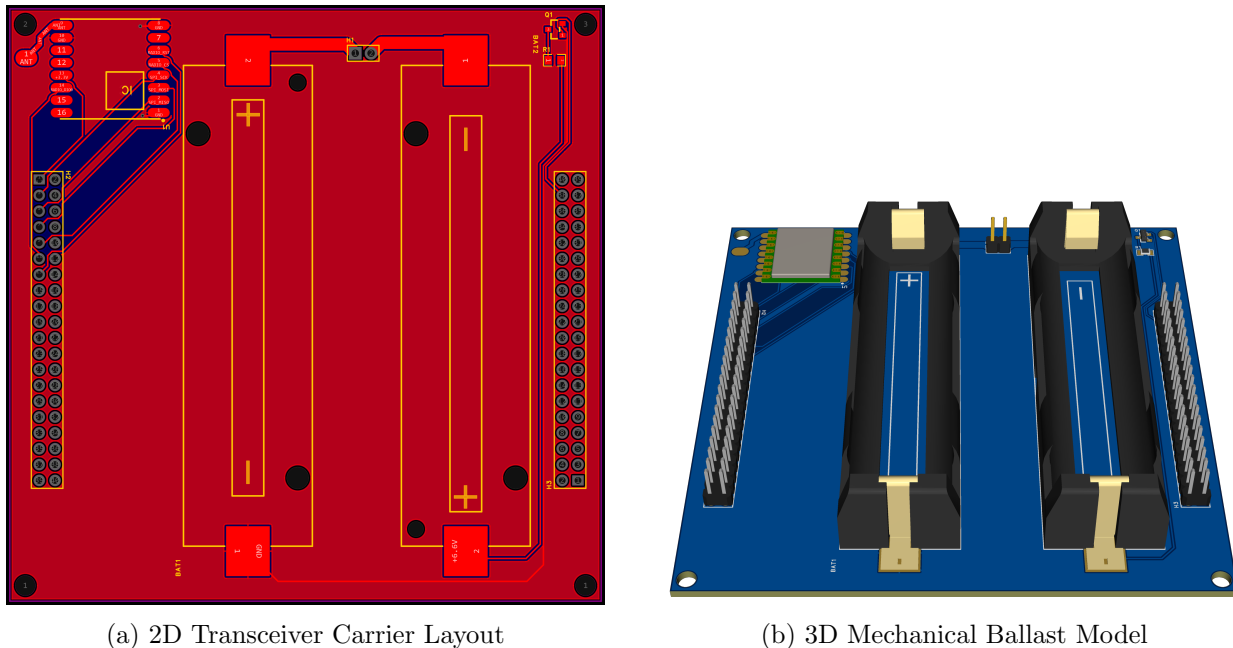


Figure 8: Hardware integration views for the Stage 5 Power Floor and UHF Telemetry module. Plot (a) shows the 2D high-speed RF trace layouts, grounding fences, and coplanar waveguide paths isolating the sub-GHz lines. Model (b) documents the physical placement of the twin 18650 Lithium Iron Phosphate (LiFePO_4) cells locked inside rigid through-hole Keystone 1042 clips, acting as a high-mass structural basement ballast, directly adjacent to the HopeRF RFM95W transceiver breakout board module footprint.

4.7 Stage 6 – Antenna Deployment Deck

Stage 6 is the final mechanical and electrical release tier positioned at the absolute base of the 1U skeleton, separated from Stage 5 via 6mm non-conductive standoffs. This layer houses the dual deployable antenna arrays, consisting of two flexible steel tape-measure elements cut to an exact quarter-wavelength dimension of 16.3 cm each to match our 437 MHz operating frequency.

Mechanically, the two spring-steel radiator arms are folded flat and wound overlapping each other inside non-conductive, 3D-printed Ultem blocks. They are bound down under tension by a single 40 lb test monofilament nylon structural constraint line drawn directly across the face of a 10-Ohm,

1W surface-mount burn resistor (2512 package, LCSC Part: C26074). When the Stage 3 computer clears its 45-minute countdown clock and initiates the antenna deployment sequence, it asserts a HIGH logic trigger to activate an IRLML2502TRPBF N-Channel MOSFET on this deck. This dumps raw battery voltage directly through the 10-Ohm resistor, creating 4.35 Watts of highly localized thermal dissipation that melts the nylon string in under 2 seconds, allowing the elastic tape measures to snap outward into a balanced, omnidirectional toroidal dipole array [4.1].

4.8 Current Assembly Progress & Mechanical Envelope Integration

The mechanical integration of the LEO-GRM nanosatellite framework represents a modular, multi-tier assembly currently utilizing a functional 4-stage engineering stack. The layout merges Stage 3 (C&DH Flight Computer Core), Stage 4 (Electrical Power System Hub), Stage 5 (High-Mass Battery Floor Ballast), and Stage 6 (Mechanical Antenna Deployment Deck) into a cohesive vertical structural column. By locking the lower 4-layer core matrix onto continuous corner M3 stainless steel threaded alignment support rods, axial structural stiffness is maximized to withstand launch vehicle axial and lateral vibration limits [4.1].

The primary objective of this current structural layout configuration is to experimentally isolate and verify the low-frequency mechanical deployment loops (Stage 6) and high-current power distribution lines (Stage 4) before integrating the delicate top-tier payload front-end sensors (Stage 1 and Stage 2) [4.1]. To support structural engineering reviews, Figure 9 documents the exact mechanical clearances, physical component positioning metrics, and volumetric space-claim envelopes across three distinct tracking facets.

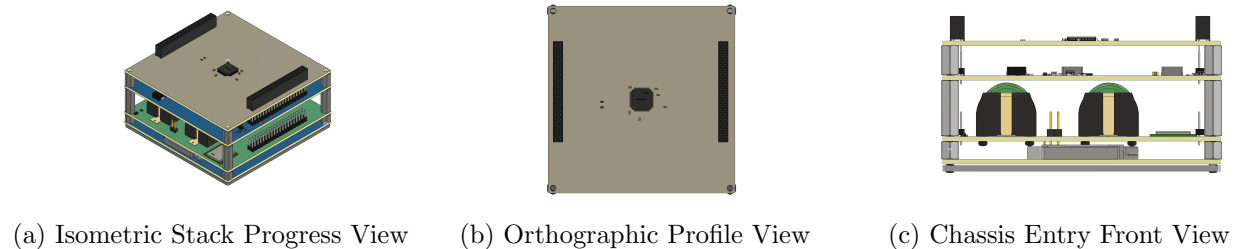


Figure 9: Autodesk Fusion 360 physical CAD assembly tracking views documenting structural integration progress as of August 11, 2026. Plots illustrate the mating clearances for the active 4-stage mechanical and electrical core layers (Stages 3, 4, 5, and 6) seated within the standardized 1U Aluminum 6061-T6 chassis guide skeleton. Panel (a) captures the isometric layer spacing, panel (b) verifies the orthogonal component profile heights, and panel (c) models the structural face alignment clearances.